Optimal Execution with Alpha

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From Liquidation to Alpha

A Primer on Optimal Execution solved the Almgren–Chriss liquidation problem [1] with no directional view: the price drift α was set to zero. Here the trader has a view. Holding a share earns an expected return $\alpha(t)$, and the case of interest is an earnings event, whose alpha arrives in a single instant. The optimal schedule turns out to be the liquidation curve plus a Green's-function response to the event. Adding the event's own jump variance then caps the position carried into the print. The last section compares the model with the round-trip version in Paleologo [2].

We assume permanent impact is linear in the trading rate, with a constant coefficient γ. In continuous time, its cost is then $\frac{1}{2}\gamma(x(T)^2 - x(0)^2)$ whatever the schedule, because it depends only on the endpoints. In a liquidation, from $x(0) = X$ to $x(T) = 0$, it equals $\frac{1}{2}\gamma X^2$, so only the temporary impact η shapes the schedule.

The Objective

The trader holds $x(t)$ shares and must go from $x(0) = X$ to $x(T) = 0$. At every instant three things happen:

- trading costs $\eta\dot{x}^2$, quadratic in speed, so trading twice as fast costs four times as much;
- holding costs risk $\lambda\sigma^2 x^2$;
- holding earns alpha $\alpha(t)x$.

The trader chooses the whole path to minimize

$$J[x] = \int_0^T \left[\eta\dot{x}(t)^2 + \lambda\sigma^2 x(t)^2 - \alpha(t)x(t)\right] \mathrm{d}t. \quad (1)$$

This is the primer's $E + \lambda V$ with the drift switched back on: every share held through a drift α earns it, which lowers the expected shortfall by $\int \alpha(t)x(t)\,\mathrm{d}t$.

The Optimality Equation

To minimize a function of many variables, set every partial derivative to zero: the first-order condition. Split $[0, T]$ into N buckets of length τ, with holdings $x_1, \ldots, x_{N-1}$ free and the endpoints $x_0 = X$ and $x_N = 0$ fixed. The objective becomes

$$J = \sum_k \left[\frac{\eta}{\tau}(x_{k-1} - x_k)^2 + \lambda\sigma^2\tau x_k^2 - \alpha_k\tau x_k\right].$$

A single holding x_k appears in exactly four places: the trade into bucket k, the trade out of it, its own risk and its own alpha. Setting its derivative to zero,

$$\frac{2\eta}{\tau}(2x_k - x_{k-1} - x_{k+1}) + 2\lambda\sigma^2\tau x_k - \alpha_k\tau = 0,$$

and dividing by $2\eta\tau$,

$$\frac{x_{k+1} - 2x_k + x_{k-1}}{\tau^2} = \kappa^2 x_k - \frac{\alpha_k}{2\eta}, \quad \kappa^2 = \frac{\lambda\sigma^2}{\eta}.$$

The left side is the change in trading speed, from $(x_k - x_{k-1})/\tau$ into bucket k to $(x_{k+1} - x_k)/\tau$ out of it, divided by τ: a discrete acceleration. As $\tau \to 0$ the first-order condition becomes, at every instant t,

$$\ddot{x} = \kappa^2 x - \frac{\alpha}{2\eta}. \quad (2)$$

This is a balance of forces. In a liquidation the trader is selling, so the trading rate $\dot{x}$ is negative, and $\ddot{x}$ is how that rate changes: $\ddot{x} > 0$ means selling slows down, and $\ddot{x} < 0$ means it speeds up. Two forces set it:

- Risk, $\kappa^2 x$, is positive and largest when the position is large. It slows the selling down over time, so the trader sells hardest while the position, and its risk, is large.
- Alpha, $-\alpha/2\eta$, is negative when $\alpha > 0$. It speeds the selling up over time, so the trader holds on now and sells later.

Each discrete condition links only x_{k-1}, x_k and x_{k+1}, so the discrete problem is a tridiagonal linear system.

No Alpha: The Liquidation Curve

With $\alpha = 0$, equation (2) is $\ddot{x} = \kappa^2 x$, solved by $e^{\pm\kappa t}$. The combination with $x(T) = 0$ and $x(0) = X$ is

$$x_{\text{base}}(t) = X\frac{\sinh\kappa(T - t)}{\sinh\kappa T}, \quad (3)$$

the continuous limit of the primer's schedule, where the discrete decay rate θ tends to κ. $1/\kappa = \sqrt{\eta/\lambda\sigma^2}$ is the time scale of the unwind.

An Earnings Event

Let all the alpha arrive at the print, $\alpha(t) = a\delta(t - t_e)$, so the trader earns $ax(t_e)$ on whatever is held through it. Integrating (2) over a vanishing window around t_e: the integral of $\ddot{x}$ is the change in speed across the window,

the integral of $\kappa^2 x$ vanishes, and the delta integrates to a, so

$$\dot{x}(t_e^+) - \dot{x}(t_e^-) = -\frac{a}{2\eta}. \quad (4)$$

The position is continuous, since a jump would need infinite speed and infinite cost. Only the trading rate jumps: for $a > 0$ the trader sells $a/2\eta$ shares per day faster after the print, so the path has a corner at t_e.

The Green's Function

Let $G(t, s)$ be the response to a unit impulse at time s with both endpoints held at zero: $\partial_t^2 G - \kappa^2 G = -\delta(t - s)$, $G(0, s) = G(T, s) = 0$. Away from s it solves the unforced equation, and gluing the two sides together at s gives

$$G(t, s) = \frac{\sinh(\kappa t_<) \sinh \kappa(T - t_>)}{\kappa \sinh \kappa T}, \quad (5)$$

where $t_<$ is the smaller of t and s, and $t_>$ the larger. Equation (2) is linear, so solutions add. The base curve (3) carries the boundary conditions and the Green's term carries the event while vanishing at both ends:

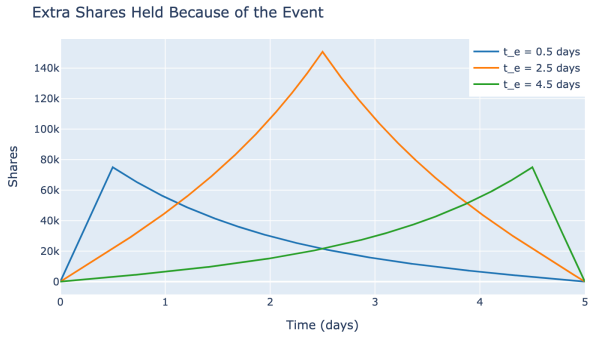
$$x(t) = x_{\text{base}}(t) + \frac{a}{2\eta} G(t, t_e). \quad (6)$$

For a general alpha the second term becomes $(2\eta)^{-1} \int_0^T G(t, s)\alpha(s)\,\mathrm{d}s$.

For $a > 0$ the trader holds extra inventory into the print and sells it off faster afterward. For $a < 0$ the trader sells ahead and arrives lighter. With the Almgren–Chriss Table 1 stock ($X = 10^6$ shares, $\sigma = 0.95$, $\eta = 2.5 \times 10^{-6}$, $T = 5$ days), $\lambda = 10^{-6}$ ($\kappa = 0.60$ per day) and a mid-horizon print worth $a = 1$ dollar per share, the trader carries 363,000 shares through it instead of 212,000.



The bump reaches only about $1/\kappa$ either side of the print, so an event far away barely affects trading today. An event near either end of the horizon has little effect, because G vanishes there and there is no time to build or unwind the extra position.



Event Risk

An earnings print also carries jump variance σ_J^2. Adding $\lambda\sigma_J^2 x(t_e)^2$ to (1), the jump condition picks up a second term,

$$\dot{x}(t_e^+) - \dot{x}(t_e^-) = \rho x_e - \frac{a}{2\eta}, \quad \rho = \frac{\lambda\sigma_J^2}{\eta}, \quad (7)$$

with $x_e = x(t_e)$. The impulse now depends on the unknown x_e, but superposition still holds:

$$x(t) = x_{\text{base}}(t) + G(t, t_e)\left(\frac{a}{2\eta} - \rho x_e\right).$$

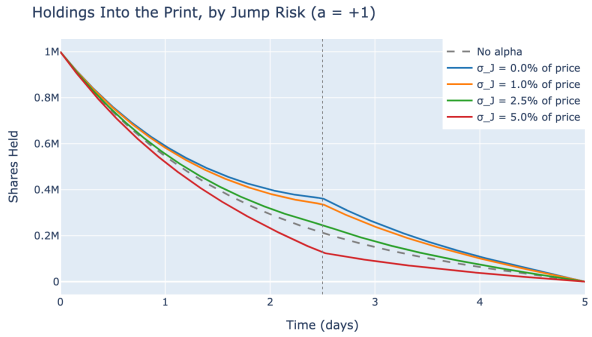
Setting $t = t_e$ and solving for x_e, with $G_{ee} = G(t_e, t_e)$,

$$x_e = \frac{x_{\text{base}}(t_e) + G_{ee}\, a/2\eta}{1 + \rho G_{ee}} = w x_{\text{base}}(t_e) + (1 - w)x^\star, \quad (8)$$

where

$$w = \frac{1}{1+\rho G_{ee}}, \quad x^\star = \frac{a}{2\lambda\sigma_J^2}.$$

$x^\star$ is the position the event alone would call for: minimizing $\lambda\sigma_J^2 x^2 - ax$. The position at the print is a blend of where the liquidation would put the trader and what the event alone wants, and the larger the jump risk, the more the event wins. When $x^\star < x_{\text{base}}(t_e)$, the trader arrives lighter than with no view at all, even though the alpha is positive. In the example, a jump of 5% of the price ($\sigma_J = 2.5$ dollars) gives $w = 0.35$ and $x^\star =$ 80,000, and the position at the print falls to 126,000 shares.



The Book, Translated

Paleologo [2] solves this model for a round trip: start flat, build a position to capture the event alpha, and unwind it ($X = 0$), with no event-variance term. His symbols map into ours as follows.

book	here
$\kappa^2/2$	η
$\rho\sigma^2/2$	$\lambda\sigma^2$
b	κ
α_0, t_0	a, t_e

With these substitutions the book's equation $\kappa^2\ddot{x} = -\alpha + \rho\sigma^2 x$ is exactly (2). Its F, G and c solve it by variation of parameters: F and G are the two running integrals of $\alpha e^{\pm b\zeta}$, and c enforces $x(T) = 0$. That is a longer route to the same Green's function. With $X = 0$ the base curve vanishes and only the Green's term of (6) remains.

Long horizon. As $T \to \infty$, $\sinh\kappa(T - t_>)/\sinh\kappa T \to e^{-\kappa t_>}$, so

$$x(t) = \frac{a}{2\eta\kappa}\sinh(\kappa t_<)e^{-\kappa t_>},$$

$$x(t_e) = \frac{a}{4\eta\kappa}(1 - e^{-2\kappa t_e}), \quad (9)$$

which are the book's long-horizon results. Before the event the trader buys ever faster, at a rate proportional to $\cosh\kappa t$. After it the position unwinds along $e^{-\kappa t}$, quickly at first and then slowly.

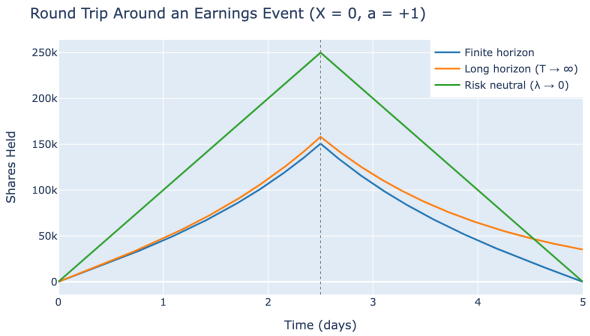
The trading rate at the event. Just before the print the buying rate is $\frac{a}{4\eta}(1 + e^{-2\kappa t_e})$, and just after it the selling rate is $\frac{a}{4\eta}(1 - e^{-2\kappa t_e})$. They differ by exactly the jump $a/2\eta$ of (4). The book says they are equal in size, which holds only when the event is far away ($\kappa t_e \gg 1$).

Risk-aversion convention. The book's problem statement prices risk as $\rho\sigma^2/2$, but its Lagrangian and its rate $b = \rho\sigma/\kappa$ use $\rho^2\sigma^2$. One convention has to be picked. With $\rho\sigma^2$, the rate is $b = \sigma\sqrt{\rho}/\kappa$.

Equation (11.9). The book's risk-neutral solution has two branches that do not meet at t_0: the left gives $\alpha_0 t_0/\kappa^2$ and the right $\alpha_0 t_0(1 - t_0)/\kappa^2$. It also mixes $T \to \infty$ with $T = 1$. The limit $\lambda \to 0$ of (5) is $G(t, s) \to t_<(T - t_>)/T$, so the position is a tent, continuous at the print:

$$x(t) = \frac{a}{2\eta}\frac{t_<(T - t_>)}{T}. \quad (10)$$

With $T = 1$ the left branch is $\alpha_0(1 - t_0)t/\kappa^2$, not $\alpha_0 t/\kappa^2$. With $T \to \infty$ the position after t_0 is flat: a risk-neutral trader never unwinds.



What the book leaves out. The book notes that conviction is strongest on the earnings date, but its risk term uses the same σ everywhere. The position it carries through the print therefore ignores the print's own variance, which is the largest risk in the trade. With $X = 0$, (8) gives

$$x_e = \frac{G_{ee}a/2\eta}{1 + \rho G_{ee}} \to x^\star = \frac{a}{2\lambda\sigma_J^2}$$

as $\sigma_J \to \infty$: the event variance caps the book's position at the event-only size, $\alpha_0/\rho\sigma_J^2$ in the book's units.

Volume time. The book measures time in volume and keeps σ and κ constant on that clock. That is an assumption about clock time: both are constant in volume time only if variance per unit clock time grows with volume, $\sigma_t^2 \propto v_t$, and impact shrinks with it, $\eta_t \propto 1/v_t$, so that $\sigma_t^2\eta_t$ is constant (appendix).

Appendix

Optimality Equation in Continuous Time

Equation (2)

The objective (1) is a functional: it assigns a number $J[x]$ to a whole path $x(t)$. Minimizing it is a problem in the calculus of variations, and the first-order condition is the Euler-Lagrange equation. For $J = \int_0^T L(x, \dot{x}, t)\,\mathrm{d}t$ it reads

$$\frac{\partial L}{\partial x} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{x}} = 0.$$

Here $L = \eta\dot{x}^2 + \lambda\sigma^2 x^2 - \alpha x$, so $\partial L/\partial x = 2\lambda\sigma^2 x - \alpha$ and $\mathrm{d}/\,\mathrm{d}t(\partial L/\partial \dot{x}) = 2\eta\ddot{x}$. Setting their difference to zero,

$$2\lambda\sigma^2 x - \alpha - 2\eta\ddot{x} = 0.$$

Dividing by 2η and writing $\kappa^2 = \lambda\sigma^2/\eta$,

$$\ddot{x} = \kappa^2 x - \frac{\alpha}{2\eta},$$

which is equation (2). The steps below derive this directly for our L, which is the general argument behind the formula.

Suppose $x(t)$ minimizes (1), and perturb it to $x + \varepsilon h$, where $h(0) = h(T) = 0$ so the endpoints stay fixed. At the minimum the derivative of $J[x + \varepsilon h]$ with respect to ε vanishes at $\varepsilon = 0$, so

$$\int_0^T \left[2\eta\dot{x}(t)\dot{h}(t) + 2\lambda\sigma^2 x(t)h(t) - \alpha(t)h(t)\right]\mathrm{d}t = 0 \quad \text{for every } h.$$

Integrating the first term by parts, and using $h(0) = h(T) = 0$ to drop the boundary term, $\int \dot{x}(t)\dot{h}(t)\,\mathrm{d}t = -\int \ddot{x}(t)h(t)\,\mathrm{d}t$, so

$$\int_0^T \left[-2\eta\ddot{x}(t) + 2\lambda\sigma^2 x(t) - \alpha(t)\right]h(t)\,\mathrm{d}t = 0 \quad \text{for every } h.$$

An integral that vanishes against every h forces the bracket to vanish. Dividing by -2η gives equation (2). Integration by parts plays the role of the discrete observation that x_k appears in two neighboring trades.

This condition is necessary. It is also sufficient here: J is a convex functional, because $J[x + \varepsilon h]$ is quadratic in ε with curvature $2\int_0^T \left[\eta\dot{h}^2 + \lambda\sigma^2 h^2\right]\mathrm{d}t \geq 0$ for $\eta > 0$ and $\lambda\sigma^2 \geq 0$. A path that satisfies (2) with the endpoint conditions is therefore the global minimum, and it is unique when $\eta > 0$.

Jump Conditions

Equations (4) and (7)

Write the event alpha as $a\delta(t - t_e)$ and include the event variance $\lambda\sigma_J^2 x(t_e)^2$. $\dot{x}$ may jump at t_e, so integrate by parts separately on $[0, t_e]$ and $[t_e, T]$:

$$\int_0^T \dot{x}(t)\dot{h}(t)\,\mathrm{d}t = h(t_e)[\dot{x}(t_e^-) - \dot{x}(t_e^+)] - \int_0^T \ddot{x}(t)h(t)\,\mathrm{d}t.$$

The first-order change in J is then

$$\int_0^T \left[-2\eta\ddot{x}(t) + 2\lambda\sigma^2 x(t)\right]h(t)\,\mathrm{d}t + h(t_e)\left[2\eta(\dot{x}(t_e^-) - \dot{x}(t_e^+)) + 2\lambda\sigma_J^2 x_e - a\right].$$

Perturbations that vanish at t_e give equation (2) away from the event. A perturbation with $h(t_e) \neq 0$ then forces the second bracket to vanish:

$$\dot{x}(t_e^+) - \dot{x}(t_e^-) = \frac{\lambda\sigma_J^2}{\eta} x_e - \frac{a}{2\eta} = \rho x_e - \frac{a}{2\eta},$$

which is (7), and (4) when $\sigma_J = 0$.

Green's Function

Equation (5)

Away from s, G solves $\partial_t^2 G = \kappa^2 G$. On each side, take the solution that already satisfies that side's boundary condition:

$$G = A \sinh \kappa t \quad (t < s), \quad G = B \sinh \kappa (T - t) \quad (t > s).$$

Two conditions at s fix A and B. G is continuous,

$$A \sinh \kappa s = B \sinh \kappa (T - s),$$

and, by (4) with a unit impulse, its slope drops by 1,

$$-B\kappa \cosh \kappa (T - s) - A\kappa \cosh \kappa s = -1.$$

Substituting $B = A \sinh \kappa s / \sinh \kappa (T - s)$ into the second condition,

$$A\kappa(\sinh \kappa s \cosh \kappa (T - s) + \cosh \kappa s \sinh \kappa (T - s)) = \sinh \kappa (T - s).$$

The bracket is $\sinh(\kappa s + \kappa(T - s)) = \sinh \kappa T$, so

$$A = \frac{\sinh \kappa (T - s)}{\kappa \sinh \kappa T}, \quad B = \frac{\sinh \kappa s}{\kappa \sinh \kappa T},$$

and both branches combine into equation (5).

Long-Horizon Limit

Equation (9)

For fixed $t_>$, as $T \to \infty$,

$$\frac{\sinh \kappa (T - t_>)}{\sinh \kappa T} = \frac{e^{-\kappa t_>} - e^{-\kappa(2T - t_>)}}{1 - e^{-2\kappa T}} \to e^{-\kappa t_>},$$

which turns (6) with $X = 0$ into the first line of (9). At $t = t_e$, $\sinh(\kappa t_e) e^{-\kappa t_e} = \frac{1}{2}(1 - e^{-2\kappa t_e})$ gives the second. Differentiating, the rate just before the print is

$$\dot{x}(t_e^-) = \frac{a}{2\eta} \cosh(\kappa t_e) e^{-\kappa t_e} = \frac{a}{4\eta}(1 + e^{-2\kappa t_e}),$$

and just after it

$$\dot{x}(t_e^+) = -\frac{a}{2\eta} \sinh(\kappa t_e) e^{-\kappa t_e} = -\frac{a}{4\eta}(1 - e^{-2\kappa t_e}).$$

Their difference is $-a/2\eta$, as (4) requires.

Risk-Neutral Limit

Equation (10)

As $\kappa \to 0$, $\sinh z \approx z$ in each factor of (5):

$$G(t,s) \approx \frac{\kappa t_< \cdot \kappa(T - t_>)}{\kappa \cdot \kappa T} = \frac{t_<(T - t_>)}{T}.$$

Both branches equal $t_e(T - t_e)/T$ at $t = s = t_e$, so the tent is continuous, and its slopes, $(T - t_e)/T$ before and $-t_e/T$ after, differ by exactly -1.

Volume Time

Let v_t be the trading volume at clock time t, V its average, and u volume time, $\mathrm{d}u = (v_t/V)\,\mathrm{d}t$. Over $\mathrm{d}t$ the price variance is $\sigma_t^2\,\mathrm{d}t = \sigma_t^2(V/v_t)\,\mathrm{d}u$, so the variance per unit volume time is $\sigma_t^2 V/v_t$. The impact cost is $\eta_t(\mathrm{d}x/\,\mathrm{d}t)^2\,\mathrm{d}t = \eta_t(v_t/V)(\mathrm{d}x/\,\mathrm{d}u)^2\,\mathrm{d}u$, so the impact per unit volume time is $\eta_t v_t/V$. Both are constant in u only if $\sigma_t^2 \propto v_t$ and $\eta_t \propto 1/v_t$, which makes $\sigma_t^2\eta_t$ constant in clock time.

Notation

symbol	meaning	units
$x(t)$	shares held at time t	shares
X	starting position ($X = 0$ for a round trip)	shares
T	trading horizon	day
η	temporary impact coefficient	\$ day/share^2
σ	price volatility	\$/(share day$^{1/2}$)
λ	risk aversion	1/\$
κ	liquidation rate, $\sqrt{\lambda\sigma^2/\eta}$	1/day
$\alpha(t)$	expected return from holding	\$/(share day)
a	alpha paid at the event	\$/share
t_e	event time	day
σ_J	jump volatility at the event	\$/share
ρ	event-risk rate, $\lambda\sigma_J^2/\eta$	1/day
$G(t,s)$	Green's function of (2)	day
x_e	position at the event, $x(t_e)$	shares
$x^\star$	event-only position, $a/2\lambda\sigma_J^2$	shares

References

[1] R. Almgren and N. Chriss, "Optimal execution of portfolio transactions," *Journal of Risk*, vol. 3, no. 2, pp. 5–39, 2000.

[2] G. A. Paleologo, *Advanced Portfolio Management: A Quant's Guide for Fundamental Investors*. Wiley, 2021.